

3 Phosphorus

Phosphorus supports energy transfer, root activity, flowering, and fruit development. It moves slowly in most soils and is strongly controlled by pH and placement.

Why it matters

- **Energy and growth.** Phosphorus is part of the energy system used in cell division, root growth, flowering, and fruit development.
- **Small amount large effect.** Plants need less P than N or K, but both deficiency and unneeded buildup can affect nutrient balance and water quality.

Tentative leaf guide

Measure	Interpretation note
Tentative avocado range	UC guidance lists less than 0.05 percent as deficient, 0.08 to 0.25 percent as adequate, and more than 0.30 percent as excess.
Interpretation	A soil test method must be matched to soil pH and local calibration. Total soil P does not equal plant-available P.

Main soil and environmental controls

Factor	What it can mean in the root zone
pH	At low pH, P can bind with iron and aluminum. At high pH, it can form less-soluble calcium compounds.
Microbes and organic acids	Some roots, fungi, and bacteria release acids or enzymes that free P locally. PGPR products are strain- and site-dependent.
Organic matter	Decomposition supplies P and may reduce some fixation. Repeated manure or compost use can also build P above crop need.
Texture and placement	P usually moves little except with eroded soil, runoff, preferential flow, or some sandy soils. Placement near active roots matters.
Water and roots	Dry soil slows diffusion to roots. Saturation, compaction, or root disease reduces root exploration and can create an apparent shortage.

Field clues and checks

- **Confirm before adding.** Compare standardized leaf results with a suitable soil P test, pH, input history, rooting, and irrigation pattern.
- **Check micronutrients.** High P can worsen zinc or iron chlorosis in susceptible settings.
- **Protect surface water.** Avoid surface buildup, erosion, runoff, and placement where irrigation water can carry P away.

DST interpretation The Phase 1 model treated phosphorus as part of a multnutrient background. P by B and Zn by B were research candidates only, not field application rules.